CSA07-CN_CO5_AT2

Pseudocode Writing Task (Algorithm Design)

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Question 1: P2P File Sharing Algorithm

The correct order of the pseudocode blocks is:

1. **START**
2. **CONNECT TO PEER**
3. **IF FILE IS AVAILABLE THEN**
4. **DOWNLOAD FILE**
5. **ENDIF**
6. **DISPLAY "File not available"**
7. **END**

⚠️ **But there is a problem:** With the given blocks, `"File not available"` will be displayed **even when the file is available**, because there is no `ELSE` block.

So, strictly using **all blocks exactly once**, the above is the only reasonable arrangement, but it is **not logically correct** for the stated scenario.

A truly correct algorithm would need:

START

CONNECT TO PEER

IF FILE IS AVAILABLE THEN

    DOWNLOAD FILE

ELSE

    DISPLAY "File not available"

ENDIF

END

That requires an **ELSE** block, which is missing from the provided blocks.

Question 2: VoIP Call Connection Algorithm

arrange the blocks in this exact order:

1. **START**
2. **CHECK INTERNET CONNECTION**
3. **IF INTERNET IS AVAILABLE THEN**
4. **CONNECT CALL**
5. **START VOICE COMMUNICATION**
6. **ELSE**
7. **DISPLAY "No Internet Connection"**
8. **ENDIF**
9. **END**

Final arrangement:

START

CHECK INTERNET CONNECTION

IF INTERNET IS AVAILABLE THEN

    CONNECT CALL

    START VOICE COMMUNICATION

ELSE

    DISPLAY "No Internet Connection"

ENDIF

END

✅ This is the **correct logical algorithm** and uses **every given block exactly once**.

Question 3: Overlay Network Message Routing Algorithm

arrange the blocks in this exact order:

1. **START**
2. **CHECK DESTINATION NODE**
3. **IF DESTINATION NODE IS AVAILABLE THEN**
4. **SEND MESSAGE DIRECTLY**
5. **ELSE**
6. **FORWARD MESSAGE THROUGH ANOTHER NODE**
7. **DISPLAY "MESSAGE SENT"**
8. **ENDIF**
9. **END**

Final arrangement:

START

CHECK DESTINATION NODE

IF DESTINATION NODE IS AVAILABLE THEN

    SEND MESSAGE DIRECTLY

ELSE

    FORWARD MESSAGE THROUGH ANOTHER NODE

DISPLAY "MESSAGE SENT"

ENDIF

END

✅ This uses **all blocks exactly once** and has **one action for each condition**.